Task Assignment & Roles

Customer Support RAG Chatbot

Project Type: Depi Graduation Project
Team Size: 5 Members
Project Duration: 11 Weeks

1 Work Distribution

The project was divided into three main parts, with team members assigned based on their expertise and areas of interest.

Part 1
Data Collection & Retrieval

Shrouk Eissa & Ahmed Elsenosy

- ✓ Data collection & exploration
- ✓ Data cleaning & preprocessing
- ✓ Exploratory Data Analysis (EDA)
- ✓ Text chunking & embedding
- ✓ ChromaDB vector database setup
- ✓ Retriever module & confidence scoring

Part 2
Generation & Backend

Youssef Ashraf & Ahmed Elsenosy

- ✓ LLM Generator (Groq API)
- ✓ Adaptive RAG pipeline
- ✓ Intent Router (CASUAL/SUPPORT)
- ✓ FastAPI REST API
- ✓ PostgreSQL database & ORM
- ✓ Authentication & security
- ✓ Voice input (Whisper)

Part 3
Frontend & MLOps

Ashraqat Effat, Fatma Shehata & Ahmed Elsenosy

- ✓ Next.js web application
- ✓ Chat interface UI
- ✓ User authentication UI
- ✓ MLflow experiment tracking
- ✓ Prometheus metrics
- ✓ Streamlit monitoring dashboard

2 Team Roles & Responsibilities

Each team member was assigned a defined role with specific responsibilities across the project milestones.

Ahmed Elsenosy Team Lead

FULL-STACK AI DEVELOPER & PROJECT LEAD

- Overall project architecture, planning, and coordination across all 3 parts
- Data pipeline: data collection, cleaning, EDA, and chunking strategy
- NLP pipeline: embedding, retrieval, and ChromaDB vector database
- LLM integration: Generator module (Groq API), Adaptive RAG pipeline, Intent Router
- Backend: FastAPI API, PostgreSQL database schema, authentication (OAuth, JWT, OTP)
- MLOps: MLflow experiment tracking, Prometheus metrics, monitoring dashboard
- Frontend: Next.js web application guidance and integration
- Testing: pipeline benchmark suite, code review, and quality assurance

Shrouk Eissa

DATA & AI DEVELOPER

- Raw dataset collection (BrownBox e-commerce Q&A)
- Exploratory Data Analysis (EDA) notebooks
- Data cleaning and preprocessing pipeline
- Text chunking and QA pair extraction
- Embedding experiments and evaluation
- ChromaDB data ingestion and validation

Youssef Ashraf

BACKEND & AI DEVELOPER

- FastAPI REST API development and endpoint design
- PostgreSQL database models and CRUD operations
- LLM Generator integration (Groq API)
- Adaptive RAG pipeline implementation
- Authentication system (Google OAuth, JWT, OTP)
- Voice input integration (Whisper speech-to-text)

Ashraqat Effat

FRONTEND DEVELOPER & MLOPS

- Next.js frontend application development
- Chat interface and conversation management UI
- User authentication pages (login, register, OAuth)

Fatma Shehata

FRONTEND DEVELOPER & MLOPS

- Next.js frontend components and pages
- UI/UX design and styling
- Frontend-backend API integration
- Streamlit monitoring dashboard development

- MLflow experiment tracking setup
- Prometheus metrics integration
- Responsive design implementation

- Configuration comparison experiments (MLflow)
- Deployment and documentation support

3 Responsibility Matrix

The matrix below maps each project milestone to the responsible team members. **R** = Responsible (primary), **C** = Contributor (supporting).

MILESTONE / TASK	AHMED ELSENOSY	SHROUK EISSA	YOUSSEF ASHRAF	ASHRAQAT EFFAT	FATMA SHEHATA
M1 Data Collection & Cleaning	R	R			
M2 Retrieval	C	R			
M3 Generation	R		R		
M4 Backend	R		R		
M5 Frontend	C			R	R
M6 MLOps	C			R	R

R = Responsible (primary owner) C = Contributor (supporting role)